Faster Sampling: Distillation

Lecture 09

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Outline

Motivation and Viewpoints

Distribution-Level Distillation

Flow-Map-Level Distillation

Takeaways

Motivation and Viewpoints

Why distillation?

Sampling from flow and diffusion models is expensive because generation is *iterative*.

- Strong training-free ODE solvers often still need about $10-20$ network evaluations (Lu et al.2022, 2023; Zheng et al.2023).
- Naive diffusion discretizations are much slower; original DDPM sampling used roughly 1000 steps (Ho et al.2020).
- **Distillation** shifts compute to training time: a slow teacher supervises a fast student (Luhman and Luhman2021; Salimans and Ho2022; Yin et al.2024; Zhou et al.2024).

Key Idea (Inference-time promise**)**

Use the teacher only during training, then sample with the student in *few steps* — ideally even *one step* (Salimans and Ho2022; Yin et al.2024; Zhou et al.2024; Song et al.2023).

The reconstruction / x_0 perspective

Under the linear-Gaussian noising model,

$$x_t \mid x_0 \sim \mathcal{N}(\alpha_t x_0,\, \beta_t^2 I_d), \qquad 0 \le t \le 1,$$

a diffusion model can be viewed as a family of denoisers.

Key Idea (Bayes + reconstruction**)**

$$x_0^*(x,t) = \mathbb{E}[X_0 \mid X_t = x] = \frac{1}{\alpha_t}\Big(x + \beta_t^2\, \nabla_x \log p_t(x)\Big).$$

If a model predicts noise $\varepsilon_\theta(x,t)$, then

$$x_{0,\theta}(x,t) = \frac{x - \beta_t \varepsilon_\theta(x,t)}{\alpha_t}.$$

So distillation can be interpreted as learning to recover x_0 — or equivalently the score / noise — in *many fewer steps* (Ho et al.2020; Song et al.2020).

Two distillation paradigms

Distribution-level

Match the student's output distribution to the teacher / data distribution.

$$\min_{\theta} \; D\big(p_0^{\theta} \,\|\, p_0^{\phi^\star}\big).$$

Examples: DMD, VSD, SiD (Yin et al.2024; Wang et al.2023; Zhou et al.2024).

Flow-map-level

Learn coarse transport maps between time marginals.

$$G_\theta(\cdot, s, t) \approx \Psi_{s\to t}.$$

Examples: KD, progressive distillation (PD) (Luhman and Luhman2021; Salimans and Ho2022).

Chronology: **KD/PD** appeared earlier; **distribution-level** methods became prominent later (Luhman and Luhman2021; Salimans and Ho2022; Wang et al.2023; Yin et al.2024; Zhou et al.2024).

A unifying flow-map view

The probability-flow ODE is

$$\frac{\mathrm{d}x_t}{\mathrm{d}t} = f_t(x_t) - \frac{1}{2} g_t^2 \nabla_x \log p_t(x_t) \;:=\; v^\star(x_t, t),$$

and its oracle flow map is

$$\Psi_{s\to t}(x_s) = x_s + \int_s^t v^\star(x_\tau, \tau)\,\mathrm{d}\tau.$$

In practice $\Psi_{s\to t}$ is unknown, so we replace it with a teacher-induced transition map

$$\mathrm{Solver}_{s\to t}(x_s; \phi^\star).$$

- Ideal generation: draw $x_1 \sim p_{\text{init}}$ and apply $\Psi_{1\to 0}$ (Song et al.2020).
- Distillation asks whether teacher transitions can supervise a faster student map (Luhman and Luhman2021; Salimans and Ho2022).
- This is the common lens behind both KD and PD (Luhman and Luhman2021; Salimans and Ho2022).

KD and the oracle objective

Knowledge distillation (KD): regress a one-step student to a long teacher rollout (Luhman and Luhman2021),

$$\mathcal{L}_{\mathrm{KD}}(\theta) = \mathbb{E}_{x_1 \sim p_{\mathrm{init}}}\Big[\| G_\theta(x_1) - \mathrm{Solver}^{\circ}_{1\to 0}(x_1; \phi^\star)\|_2^2\Big].$$

A more general ideal objective is

$$\mathcal{L}_{\mathrm{oracle}}(\theta) = \mathbb{E}_{(s,t)\sim\pi}\mathbb{E}_{X_s\sim p_s}\Big[w(s,t)\, d\big(G_\theta(X_s, s, t), \Psi_{s\to t}(X_s)\big)\Big].$$

Key Idea (Unification)

- **KD**: put all mass of π on $(s,t) = (1,0)$ (Luhman and Luhman2021).
- **PD**: use many *local* pairs (s,t) and match short teacher fragments (Salimans and Ho2022).

Distribution-Level Distillation

Distribution-level setup and the VSD objective

Let the student generator be $\hat{x}_0 = G_\theta(z)$ with $z \sim p_{\text{init}}$, so $\hat{x}_0 \sim p_0^\theta$. Using the same Gaussian noising process as the teacher,

$$\hat{x}_t = \alpha_t G_\theta(z) + \beta_t \varepsilon, \qquad z \sim p_{\text{init}},\ \varepsilon \sim \mathcal{N}(0, I_d), \qquad \hat{x}_t \sim p_t^\theta.$$

Variational score distillation (VSD)

$$\mathcal{L}_{\text{VSD}}(\theta) = \mathbb{E}_{t \sim p(t)}\Big[\omega(t)\, D_{\text{KL}}\big(p_t^\theta \,\|\, p_t\big)\Big].$$

Matching *noisy marginals across time* forces the student distribution to align with the teacher distribution (Wang et al.2023).

KL gives a score-difference gradient

Proposition (VSD gradient)

For $\hat{x}_t = \alpha_t G_\theta(z) + \beta_t \varepsilon$,

$$\nabla_\theta \mathcal{L}_{\mathrm{VSD}}(\theta) = \mathbb{E}_{t,z,\varepsilon}\Big[\omega(t)\,\alpha_t\,\big(\nabla_x \log p_t^\theta(\hat{x}_t) - \nabla_x \log p_t(\hat{x}_t)\big)^\top \nabla_\theta G_\theta(z)\Big].$$

- **Teacher score** $\nabla_x \log p_t(x)$: provided by a pretrained diffusion model.
- **Student score** $\nabla_x \log p_t^\theta(x)$: depends on the current generator and must be estimated.

Key Idea (Why KL is special**)**

KL yields a **clean score-difference signal**. Generic f-divergences or Renyi divergences usually introduce unknown density-ratio terms (Wang et al.2023; Nowozin et al.2016).

Bilevel training: estimate the student score

Because $\hat{x}_t \mid \hat{x}_0$ is Gaussian,

$$\nabla_x \log p_t(\hat{x}_t \mid \hat{x}_0) = -\frac{\hat{x}_t - \alpha_t \hat{x}_0}{\beta_t^2}.$$

This lets us train an auxiliary score network $s_\zeta(x, t)$ on student-generated samples via denoising score matching:

$$\mathcal{L}_{\mathrm{DSM}}(\zeta; \theta) = \mathbb{E}_{t,z,\varepsilon}\left[\left\| s_\zeta(\hat{x}_t, t) + \tfrac{\hat{x}_t - \alpha_t \hat{x}_0}{\beta_t^2} \right\|_2^2\right].$$

Then update the generator using the approximate gradient

$$\nabla_\theta \mathcal{L}_{\mathrm{VSD}}(\theta) \approx \mathbb{E}_{t,z,\varepsilon}\left[\omega(t)\, \alpha_t \left(s_\zeta(\hat{x}_t, t) - s_{\mathrm{teacher}}(\hat{x}_t, t)\right)^\top \nabla_\theta G_\theta(z)\right].$$

So training alternates between **score estimation** (update ζ) and **generator update** (update θ) (Song et al.2020; Wang et al.2023).

VSD training recipe and why it works

Practical loop

1. Sample t, latent z, and Gaussian noise ε.
2. Form $\hat{x}_0 = G_\theta(z)$ and $\hat{x}_t = \alpha_t \hat{x}_0 + \beta_t \varepsilon$.
3. Update s_ζ with DSM on student samples.
4. Update G_θ using the score-difference signal.

Key Idea (Why this works)

If student and teacher scores match over time, then $p_t^\theta \approx p_t$. Because Gaussian noising is injective, this propagates back to $p_0^\theta \approx p_0$ (Wang et al.2023).

Beyond KL and an important 3D-generation application

Beyond KL. Replacing D_{KL} by a generic f-divergence or Rényi divergence usually leads to expectations weighted by unknown density ratios p_t^θ / p_t. In practice this often means a critic / discriminator or explicit ratio estimation (Nowozin et al.2016; Wang et al.2023).

VSD for 3D generation. Let θ parametrize a 3D scene and let a differentiable renderer produce

$$\hat{x}_0 = \mathcal{R}(\theta), \qquad \hat{x}_t = \alpha_t \mathcal{R}(\theta) + \beta_t \varepsilon.$$

A score-alignment objective is

$$\mathcal{L}_{\mathrm{VSD}}^{\mathrm{3D}}(\theta) = \frac{1}{2}\mathbb{E}_{t,\varepsilon}\Big[\omega(t)\|s_\zeta(\hat{x}_t, t) - s_{\mathrm{teacher}}(\hat{x}_t, t \mid c)\|_2^2\Big].$$

Using a stop-gradient surrogate gives the practical update

$$\nabla_\theta \tilde{\mathcal{L}}_{\mathrm{VSD}}^{\mathrm{3D}}(\theta) = \mathbb{E}\Big[\omega(t)\alpha_t\big(s_\zeta - s_{\mathrm{teacher}}\big)^\top \nabla_\theta \mathcal{R}(\theta)\Big].$$

Setting $s_\zeta \equiv 0$ recovers the familiar **SDS** gradient used in text-to-3D pipelines (Wang et al.2023).

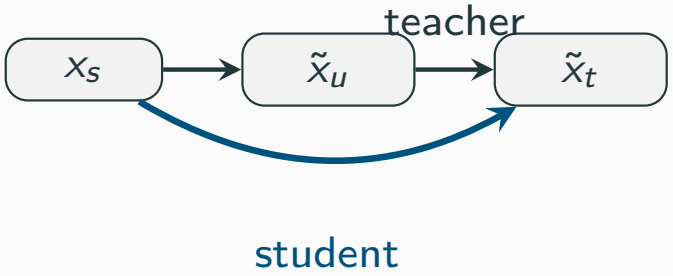
Flow-Map-Level Distillation

Progressive distillation starts from a DDIM map

For deterministic samplers, a DDIM transition from time s to t using an x_0-predictor $\hat{x}_0(\cdot, s)$ is (Song et al.2022; Salimans and Ho2022)

$$\mathrm{DDIM}_{s\to t}(x_s; \hat{x}_0) = \frac{\beta_t}{\beta_s} x_s + \alpha_s \Big(\frac{\alpha_t}{\alpha_s} - \frac{\beta_t}{\beta_s} \Big) \hat{x}_0(x_s, s).$$

- PD distills *two teacher DDIM steps into one student step* (Salimans and Ho2022).
- Fix three times $s > u > t$.
- Teacher produces $\tilde{x}_u$ then $\tilde{x}_t$.
- Student should jump directly from x_s to $\tilde{x}_t$.



Two teacher steps into one student target

Define a pseudo-target $\tilde{x}$ so that a single DDIM step from s to t matches the teacher's two-step output:

$$\tilde{x}_t = \mathrm{DDIM}_{s\to t}(x_s; \tilde{x}) = \frac{\beta_t}{\beta_s} x_s + \alpha_s \Big(\frac{\alpha_t}{\alpha_s} - \frac{\beta_t}{\beta_s} \Big) \tilde{x}.$$

Solving for $\tilde{x}$ yields the closed-form target

$$\tilde{x} = \frac{\beta_s}{\alpha_t \beta_s - \alpha_s \beta_t} \tilde{x}_t - \frac{\beta_t}{\alpha_t \beta_s - \alpha_s \beta_t} x_s.$$

The student denoiser $f_\theta(x_s, s)$ is then trained to regress to this pseudo-target.

PD objective and training pipeline

On a discrete grid $t_0 = 1 > t_1 > \cdots > t_N = 0$, progressive distillation uses

$$\min_{\theta} \mathbb{E}_k \mathbb{E}_{x_{t_k} \sim p_{t_k}} \left[w(\lambda_{t_k}) \, \| f_\theta(x_{t_k}, t_k) - \tilde{x}^{(k)} \|_2^2 \right], \qquad \lambda_t = \log(\alpha_t / \beta_t).$$

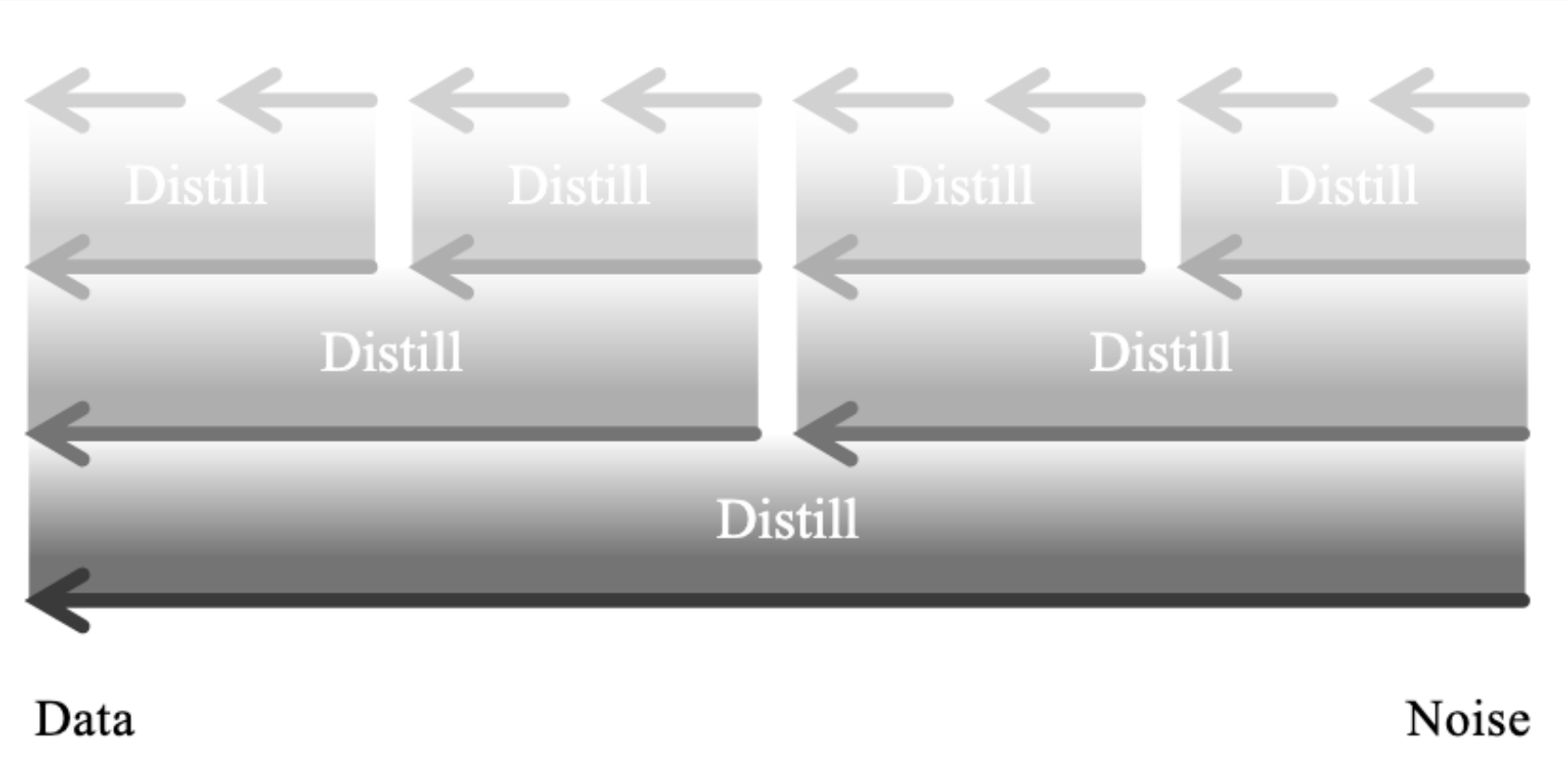
Key Idea (Why it is progressive)

1. Train a student so one step matches two teacher steps.
2. Replace the teacher by the student.
3. Keep every other time point and repeat.

Each round halves the number of sampling steps:

$$N \to N/2 \to N/4 \to \cdots$$

Progressive distillation: An graph illustration



PD as local semigroup matching

The oracle flow maps satisfy the semigroup property

$$\Psi_{s\to t} = \Psi_{u\to t} \circ \Psi_{s\to u}.$$

Progressive distillation enforces this *locally* by requiring one long student step to match two short teacher steps:

$$\mathbb{E}_s \mathbb{E}_{x_s \sim p_s} \left\| G_\theta(x_s, s, s - 2\Delta s) - \mathrm{Solver}_{s-\Delta s \to s-2\Delta s}\big(\mathrm{Solver}_{s\to s-\Delta s}(x_s)\big) \right\|_2^2.$$

So PD never needs the full teacher rollout as a target — it only matches **short teacher fragments** while recursively coarsening the grid (Salimans and Ho2022).

Practical extensions of flow-map distillation

- **Other solvers:** the DDIM pseudo-target is analytic because the update is affine in the x_0-prediction. For more general solvers, one can still match transition maps directly (Song et al.2022; Salimans and Ho2022).
- **Stochastic samplers:** freeze the random seed per example so the teacher transition becomes deterministic during regression (Luhman and Luhman2021; Salimans and Ho2022).
- **Classifier-free guidance:** per-step CFG doubles compute. A practical recipe is (Ho and Salimans2022; Meng et al.2023):
  1. distill guided and unguided predictions into a single network that takes guidance weight ω as input;
  2. then apply PD on that single-network teacher to reduce the number of steps.

Result: a sampler with *one network call per step* and a very small number of steps.

Takeaways

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- Distillation is the main **training-based** route to fewer function evaluations (Luhman and Luhman2021; Salimans and Ho2022; Yin et al.2024; Zhou et al.2024; Song et al.2023).
- **Distribution-level methods** (e.g. VSD) match teacher and student distributions; with KL, the update is a clean *score-difference* signal (Wang et al.2023; Zhou et al.2024).
- **Flow-map-level methods** (KD / PD) learn transport maps directly; PD is powerful because it recursively halves the number of sampling steps (Luhman and Luhman2021; Salimans and Ho2022).
- VSD also supports 3D generation, while PD combines naturally with guidance distillation for practical text-to-image systems (Wang et al.2023; Meng et al.2023).

Big picture

more teacher-guided training $\Longrightarrow$ **much cheaper inference.**

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